

# Compact Spherical Reactor Design for Proton–Electron Capture Chain Energy Extraction: Multi-Layer Architecture with Thermoelectric Output

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## Abstract

We present a compact spherical reactor design implementing the PECCNAL proton–electron capture chain reaction for direct electricity generation. The design employs a multi-layer architecture: a central reaction chamber containing closely-spaced nickel electrode arrays in deuterium oxide ( $D_2O$ ) with dissolved Lithium-6, surrounded by a copper thermal conductor layer, and an outer array of thermoelectric (Peltier) modules converting thermal output to electricity. The electrode array consists of alternating nickel and stainless steel plates spaced 1–2 mm apart, maximizing reaction surface area while minimizing proton travel distance. Preliminary scaling analysis suggests a 15 cm diameter prototype could produce measurable electrical output detectable with standard instrumentation.

**Keywords:** compact reactor, thermoelectric, peltier, nickel electrodes, spherical design,  $D_2O$ , prototype

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## 1 Introduction

Previous papers in this series have established the theoretical basis for PECCNAL reactions. Here we present a concrete engineering design for a compact prototype reactor intended for laboratory demonstration and calorimetric verification.

## 2 Reactor Architecture

### 2.1 Layer Structure

Total diameter:  $\approx 190$  mm. Total mass:  $\approx 2.5$  kg.

### 2.2 Electrode Array

The electrode array consists of:

Table 1: Reactor layer specifications

| Layer      | Material                                                              | Function             | Thickness  |
|------------|-----------------------------------------------------------------------|----------------------|------------|
| Core       | Ni/SS electrodes + D <sub>2</sub> O + Li <sub>2</sub> CO <sub>3</sub> | Reaction             | 60 mm dia. |
| Thermal    | Copper                                                                | Heat conduction      | 5 mm       |
| Insulation | PETG polymer                                                          | Electrical isolation | 2 mm       |
| Output     | TEC1-12706 Peltier array                                              | Heat→electricity     | 4 mm       |
| Shield     | Polyethylene                                                          | Neutron absorption   | 20 mm      |
| Housing    | Aluminium                                                             | Structural           | 3 mm       |

- 50 nickel plates (cathode, −), 60 mm × 60 mm × 0.5 mm
- 50 stainless steel plates (anode, +), same dimensions
- Plate spacing: 1 mm
- Total electrode surface area:  $50 \times 2 \times 36 \text{ cm}^2 = 3600 \text{ cm}^2$
- PETG polymer spacers (3D-printed) maintaining alignment

## 2.3 Electrolyte

D<sub>2</sub>O with 5% dissolved Li<sub>2</sub>CO<sub>3</sub> by mass:

- Volume:  $\approx 100 \text{ mL}$
- Li-6 content: 7.5% of Li mass (natural abundance)
- Conductivity enhanced by Li<sub>2</sub>CO<sub>3</sub> dissociation

## 3 Energy Flow

$$\text{Electrical input} \xrightarrow{\text{electrolysis}} \text{D}^+ \text{ ions} \xrightarrow{\text{PECCNAL}} \text{Heat} \xrightarrow{\text{Peltier}} \text{Electrical output} \quad (1)$$

Peltier efficiency at  $\Delta T = 10 \text{ K}$ :

$$\eta_{\text{Peltier}} \approx 5\text{--}8\% \quad (2)$$

## 4 Measurement Protocol

1. Measure input power:  $P_{\text{in}} = V \times I$  (multimeter)
2. Measure output power:  $P_{\text{out}} = V_{\text{Peltier}} \times I_{\text{Peltier}}$
3. Control experiment: replace D<sub>2</sub>O with H<sub>2</sub>O
4. Excess power:  $\Delta P = P$

## 5 Manufacturing

- Electrode spacers: 3D-printed PETG (Elegoo Neptune 4)
- Copper housing: cut from sheet stock, soldered
- Peltier modules: TEC1-12706, commercially available
- Total estimated cost: 200–400 USD

## 6 Conclusion

A compact 190 mm spherical reactor implementing the PECCNAL mechanism is described, using 100 alternating Ni/SS electrode plates in  $\text{D}_2\text{O}/\text{Li}_2\text{CO}_3$  electrolyte. The design is manufacturable with 3D printing and standard electronic components at low cost. Thermoelectric output and calorimetric excess heat measurement provide two independent verification channels for the PECCNAL reaction.

## References

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